Technical report: Discerning Archaeological Cereals Using Stable Isotopes

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Repository: github.com/dplauto-cpu/Carpology

1. Introduction

This project was carried out as part of an assessed practical component of a Machine Learning course. While the focus is not on carpological research, datasets from academic research have been used, and basic guidance has been provided by a member of the Research Group **isoTOPIK·Lab** at the *University of Burgos* (UBU). This is a preliminary study that is expected to be revisited at some point from a purely archaeological perspective.

1.1 Archaeological Context

The domestication of cereals (wheat and barley) during the Neolithic revolution transformed human societies. However, identifying charred archaeological seeds to species level is seriously challenging due to degradatation and carbonisation. Stable isotope analysis ($\delta 13C$ and $\delta 15N$) offers an alternative: $\delta 13C$ indicates water stress, while $\delta 15N$ reflects aridity or manuring intensity.

1.2 Research Question

Is it possible to train a ML model to distinguish barley vs wheat isotopes, and integrate the data into a coherent geographical and chronological archaeological framework?

1.3 Objectives

- **Primary:** Build a supervised model to classify barley vs wheat
- **Secondary:** Identify isotopic clusters using unsupervised learning
- **Tertiary:** Create interactive maps for archaeological interpretation

2. Dataset

2.1 Source and Composition

The dataset `Med_plants.csv` (1341 records, 20 fields) merges:

- **MAIA database** (Mediterranean Archive of Isotopic dAta)
- **IsoTOPIKLab dataset** (University of Burgos)
- 1147 samples of barley and wheat, the rest of plants unused in the ML

2.2 Target Variable Distribution

Class	Count	Percentage
Barley	586	51.1%
Wheat	561	48.9%

2.3 Main Features

Feature	Type	Description
`IRMS_d13C_Collagen`	Continuous	Carbon isotope (‰)
`d15N_Collagen`	Continuous	Nitrogen isotope (‰)
`Latitude_N`	Continuous	Latitude (decimal degrees)
`Longitude_E`	Continuous	Longitude (decimal degrees)
`Chronological_Period_clean`	Categorical	Neolithic, Bronze Age, Iron Age
`Mediterranean_Basin`	Categorical	Western, Central, Eastern, Undefined

Cultural Period	Mean date (BP)	BCE/CE range (approx.)
Neolithic	≥ 8000	≥ 6000 BCE
Neolithic_Late	6000 – 7999	4000 – 5999 BCE
Chalcolithic	5200 – 5999	3200 – 3999 BCE
Early_Bronze	4200 – 5199	2200 – 3199 BCE
Middle_Late_Bronze	3500 – 4199	1500 – 2199 BCE
Iron_Age_Early	3000 – 3499	1000 – 1499 BCE
Iron_Age_Late	2400 – 2999	500 – 999 BCE
Iron_Roman	1700 – 2399	500 BCE – 250 CE

Region	Longitude (decimal degrees)	Latitude (decimal degrees)
General limits	-10 ≤ lon ≤ 42	25 ≤ lat ≤ 46
Western_Med	-10 ≤ lon ≤ 11	25 ≤ lat ≤ 46
Central_Med	11 < lon ≤ 23	25 ≤ lat ≤ 46
Eastern_Med	23 < lon ≤ 42	25 ≤ lat ≤ 46

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3. Methodology

3.1 Preprocessing

Steps applied

1. Filter: keep only 'Hordeum vulgare' and 'Triticum dicoccum'
2. Encode categorical variables (LabelEncoder)
3. Scale numerical features (StandardScaler)
4. No missing values in key features

Libraries:

In total there are **3 main packages** to install: `scikit-learn`, `xgboost`, and the standard data science libraries (`numpy`, `pandas`, `matplotlib`, `seaborn`, `joblib`).

Standard libraries

- `numpy`
- `pandas`
- `matplotlib` (`matplotlib.pyplot`)
- `seaborn`
- `joblib`

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scikit-learn (`sklearn`)

- `sklearn.cluster` → `KMeans`
- `sklearn.compose` → `ColumnTransformer`
- `sklearn.ensemble` → `RandomForestClassifier`, `RandomForestRegressor`
- `sklearn.linear_model` → `LogisticRegression`
- `sklearn.metrics` → `accuracy_score`, `classification_report`, `confusion_matrix`, `roc_curve`, `roc_auc_score`, `mean_squared_error`, `r2_score`, `mean_absolute_error`
- `sklearn.model_selection` → `train_test_split`, `GridSearchCV`, `cross_val_score`, `StratifiedKFold`
- `sklearn.naive_bayes` → `GaussianNB`
- `sklearn.pipeline` → `Pipeline`
- `sklearn.preprocessing` → `StandardScaler`, `LabelEncoder`, `OneHotEncoder`
- `sklearn.svm` → `SVC`

XGBoost

- `xgboost` → `XGBClassifier`

3.2 Dataset Creation Process

The final dataset `Med_Plants.csv` (1341 records, 20 fields) was created through a multi-step pipeline:

Step 1: Data integration

- **MAIA database** (2,323 samples, 93 columns): extracted relevant columns including `Entry_ID`, `Site_Name`, `IRMS_d13C_Collagen`, `d15N_Collagen`, `Latitude_N`, `Longitude_E`, `Genus`, `Species`, `Cultural_Horizon`, `Absolute_Chronology`
- **Trigo_cebada dataset** (3,344 samples, 17 columns): extracted `site`, `latitud`, `longitud`, `cereal`, `taxon`, `Cultural period`, `d13C`, `d15N`
- **Merge strategy:** Left join on `Entry_ID` where available; for remaining samples, approximate join by coordinates (KDTree with 0.2° tolerance, ~22 km)

Step 2: Initial cleaning

- Removed samples without both isotopic values (δ13C and δ15N)
- Result: 1,346 samples retained (42% removed due to missing isotopes)

Step 3: Feature engineering

- **Chronological periods:** Extracted from `Absolute_Chronology` using regex patterns, converted to BP dates, then assigned to 8 standardised periods (Neolithic to Iron Age). Dates were calibrated using the conversion: BP = BCE + 1950.
- **Mediterranean basins:** Assigned based on longitude thresholds (Western: lon ≤ 11°, Central: 11° < lon ≤ 23°, Eastern: 23° < lon ≤ 42°)
- **Cereal classification:** Created `Cereal_Type` column distinguishing barley, wheat, broomcorn millet, oat, and non-cereals
- **Pulse classification:** Created `Pulse_Type` for legumes (lentil, faba bean, grass pea, pea, vetch) and other crops (olive, flax, pistachio)
- **Unified site names:** Combined `Site_Name` (MAIA) and `site` (trigo_cebada) into `Archaeological_Site` (98.7% coverage)

Step 4: Final filtering

- Removed 5 samples with undefined Mediterranean basin
- **Final dataset:** 1,341 samples × 20 columns

3.3 Exploratory Data Analysis (EDA)

The following analyses were conducted:

Isotopic distributions by period:

- Boxplots of δ13C and δ15N for each chronological period revealed a progressive trend toward more positive δ13C (increasing water stress) from Neolithic to Iron Age
- δ15N showed greater variability, with higher values in the Bronze Age suggesting possible manuring intensification

Isotopic scatter by period:

- Overlapping isotopic ranges between barley and wheat confirmed that isotopes alone cannot perfectly separate the two species
- Neolithic samples clustered in the δ13C range of -26‰ to -22‰, while Iron Age samples extended to -20‰

Geographical patterns:

- Western Mediterranean samples showed more negative δ13C (wetter conditions)
- Eastern Mediterranean samples showed more positive δ13C (drier conditions) and higher δ15N (greater aridity)

Correlations:

- Weak negative correlation between δ13C and δ15N (-0.21), suggesting that water stress and aridity/ manuring are partially independent agricultural variables

3.4 Unsupervised Learning: K-means Clustering

Objective: Identify natural groupings of samples based only on isotopic values (δ13C, δ15N)

Method:

- K-means algorithm with k=4 (determined by elbow method and silhouette analysis)
- Features scaled with StandardScaler prior to clustering

Cluster interpretation:

Cluster	δ13C (‰)	δ15N (‰)	n	Interpretation
0	-24.5	4.2	408	Good water availability, low aridity
1	-22.1	6.8	357	Moderate water stress
2	-20.3	9.5	212	High water stress, arid soils
3	-19.8	11.2	127	Extreme aridity / manuring

Geographical distribution of clusters:

- Cluster 0 (optimal conditions) dominated Western Mediterranean
- Cluster 3 (extreme stress) concentrated in Eastern Mediterranean
- This confirms that environmental gradients strongly influence isotopic signatures

3.5 Supervised Learning: Model Design and Comparison

Features selected for supervised models

Feature	Type	Encoding
IRMS_d13C_Collagen	Continuous	Scaled (StandardScaler)

Feature	Type	Encoding
d15N_Collagen	Continuous	Scaled (StandardScaler)
Latitude_N	Continuous	Scaled (StandardScaler)
Longitude_E	Continuous	Scaled (StandardScaler)
Chronological_Period_clean	Categorical	LabelEncoder
Mediterranean_Basin	Categorical	LabelEncoder

Target variable: `Cereal_Type` (barley = 1, wheat = 0)

Best-results Models evaluated

1. **Random Forest (base)** - n_estimators=100, random_state=42
2. **XGBoost** - n_estimators=100, eval_metric='logloss'
3. **Random Forest (balanced)** - class_weight='balanced'

Validation strategy: 80/20 stratified train-test split

Results comparison

Model	Accuracy	AUC	Top 1 Feature	Top 2 Feature	Top 3 Feature
Random Forest (base)	74.35%	0.825	δ13C (42%)	δ15N (29%)	lat (12%)
XGBoost	74.35%	0.813	lat (28%)	lon (22%)	δ13C (19%)
Random Forest (balanced)	**74.78%**	**0.824**	δ13C (41%)	δ15N (28%)	lat (13%)

Model selection: Random Forest with class balancing

Why Random Forest over XGBoost?

- RF gives **71% combined importance to isotopes**, while XGBoost prioritises geography (44%)
- Isotopic relationships (δ13C × δ15N) are **non-linear and interactive** – RF's ensemble of parallel trees captures these interactions naturally
- XGBoost's sequential boosting optimises for **binary spatial divisions** (latitude/longitude thresholds), which is less relevant for this research question

Why class balancing?

- The dataset is slightly imbalanced (51.1% barley vs 48.9% wheat)
- `class_weight='balanced'` increases penalty for misclassifying the minority class (wheat)
- Result: accuracy improved by **+0.43%** with stable AUC

Confusion matrix (RF balanced):

Actual \ Predicted	Wheat	Barley
Wheat	148	48
Barley	43	171

Interpretation: The model confuses wheat as barley slightly more often than the reverse, possibly due to isotopic overlap in certain environmental conditions.

Hyperparameter tuning (GridSearchCV)

Grid search was performed for Random Forest with the following parameters:

- `n_estimators`: [50, 100, 200]
- `max_depth`: [10, 20, None]
- `min_samples_split`: [2, 5, 10]

Best parameters: n_estimators=100, max_depth=20, min_samples_split=2 (no significant improvement over base model)

3.6 Feature Importance Analysis

Key insight: The model does not rely on a single feature but combines multiple sources of information:

- **δ13C (41%):** Primary predictor – captures water stress, the main environmental variable differentiating cultivation conditions
- **δ15N (28%):** Secondary predictor – reflects soil aridity or manuring intensity
- **Latitude (13%):** Geographic proxy for climatic zone
- **Longitude (10%):** Proxy for Mediterranean sub-basin
- **Period (5%):** Chronological changes in agricultural practices
- **Basin (3%):** Regional agricultural traditions

This distribution confirms that **isotopic signals are the primary drivers of classification**, with geography and chronology providing complementary context.